1

(a): Foreign direct investment is defined as the investment (movement of money) into firms/businesses within a country from outside that country (often caused by removal of barriers to trade).

(ii) Unemployment rate is defined as the relative change in unemployment over a given amount of time. Calculated by: $\frac{\text{Total unemployment}}{\text{previous unemployment}} \cdot 100$

(b)(i) Annual rate of inflation: $\frac{\text{new CPI} \cdot 100}{\text{old CPI}} - 100$

2016: 1.37 1152% 2017: 1.4427412% 2018: 0.8% 2019: N/A

(ii) Disinflation set in during 2018 to 2019

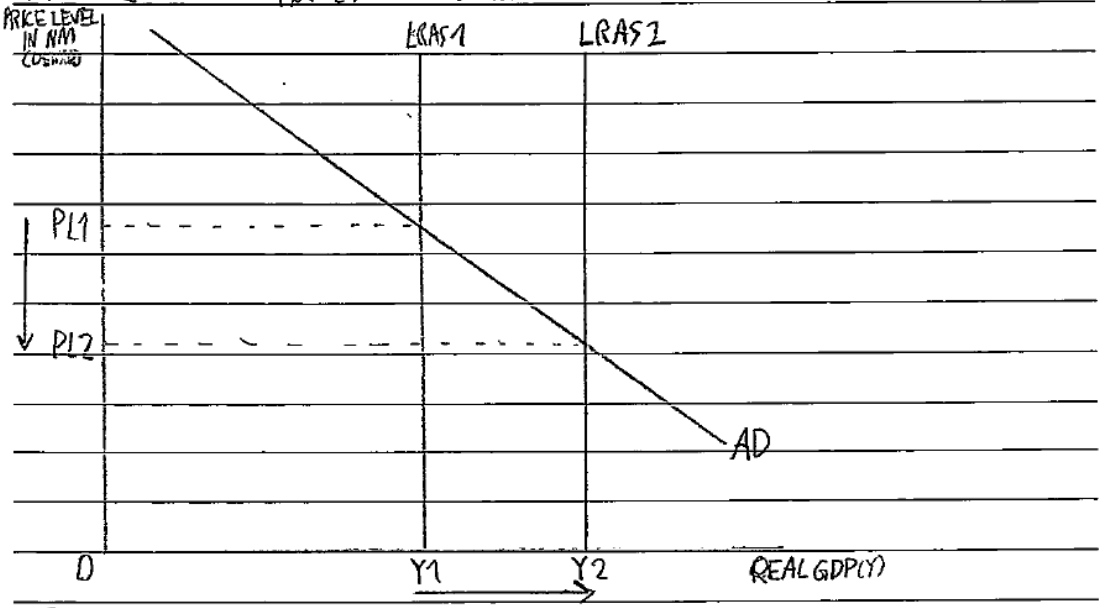
[2018]

(iii) Balance of trade in 2018 = X − M

= -1.99 billion $

(c) The likely impact of North Macedonia's supply-side policies on its full employment level of output is an increase as seen in figure 1.

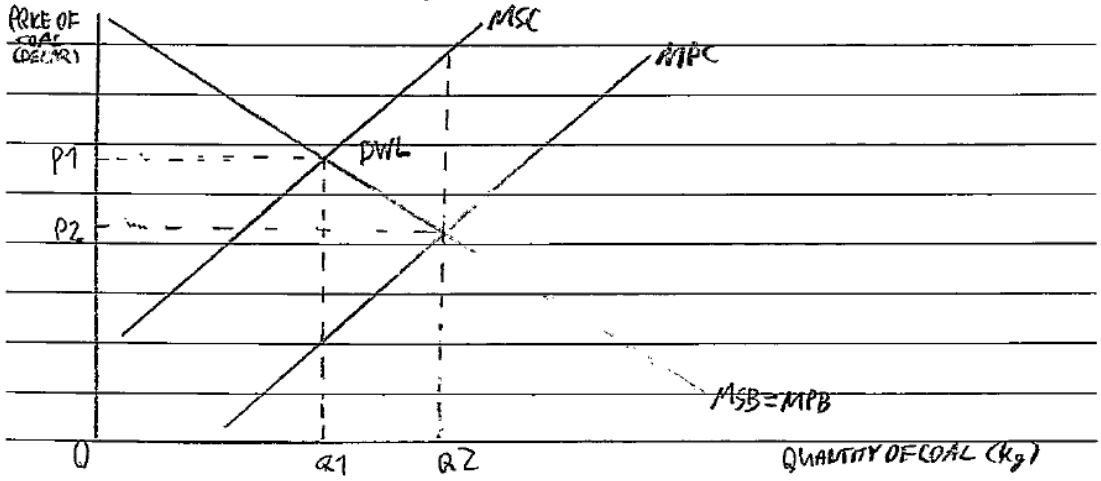
FIGURE 1. DIAGRAM OF SHIFT IN LONG-RUN AGGREGATE SUPPLY IN NORTH MACEDONIA



Supply side policies impact the aggregate supply of a country. The text makes reference to a combination of factors relating to "access to education", "training for workers" and improved "infrastructure" which are expected to increase efficiency". The improvement of supply-side factors is reflected on the diagram (fig. 1) in the shift of LRAS from LRAS1 to LRAS2. The projected improvement in "efficiency" and on the "full employment level of output" is shown as the increase in real GDP($G+I+C+(X-M)$) from Y1 to Y2. What is not mentioned in the text but may materialize is a decrease in price level from PL1 to PL2.

How dependence on coal in North-Macedonia (NM) may become a negative externality of production, in terms of producing energy, is shown in figure 2.

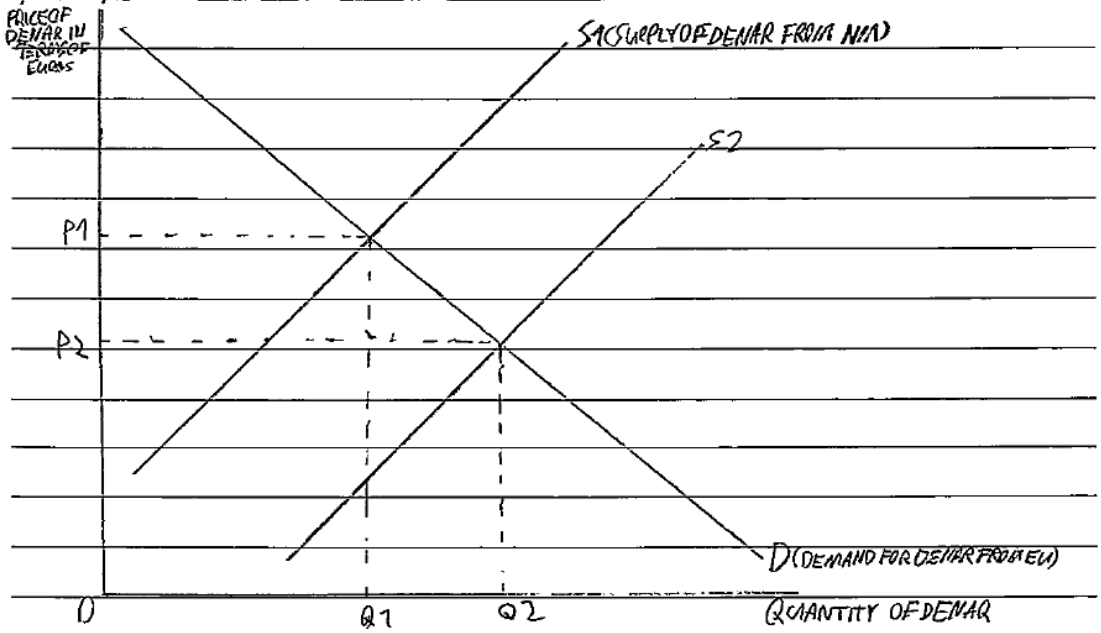
FIGURE 2. COAL USE IN NM AS NEGATIVE EXTERNALITY OF PRODUCTION



A negative externality of production is any production of a good which has a negative impact on a third party. NM's dependence on coal in energy production, causes "air pollution" and fails to follow the EU's standard. The market failure is shown as marginal social cost outweighs marginal private cost (MSC > MPC). The given scenario means that coal is used too much as seen in the change from Q1 to Q2. The externality creates a deadweight welfare loss, hence market failure. The use of coal for energy is unsustainable and represents a misallocation of resources.

(e) The efforts of the ~~the~~ NM's central bank to prevent "the denar... from appreciating against the Euro" can be observed in figure 3.

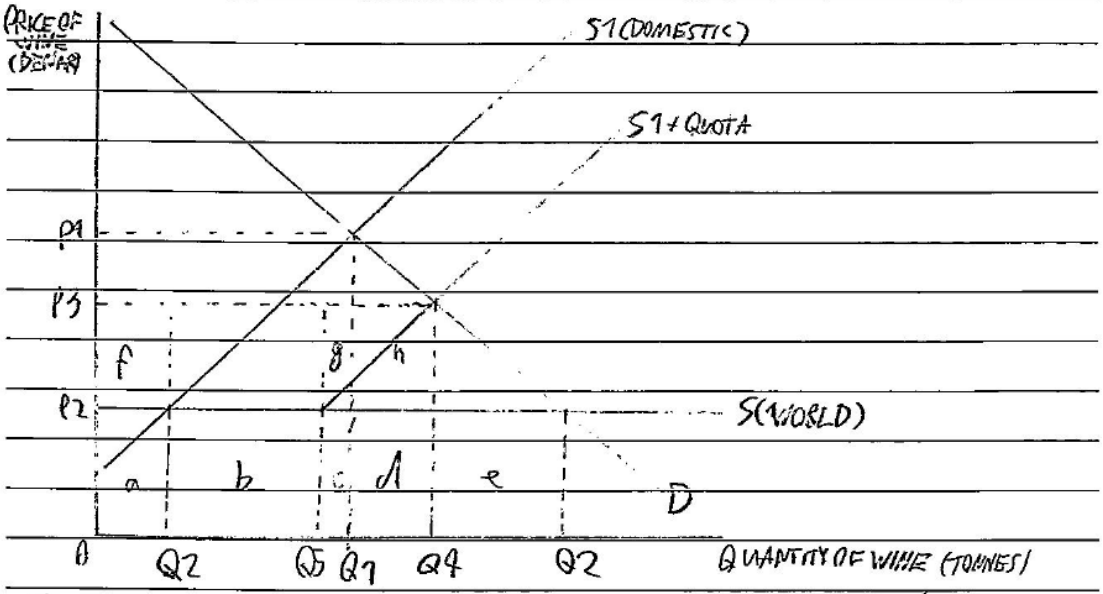
FIGURE 3. EXCHANGE RATE DIAGRAM OF DENAR IN TERMS OF EURO



In order of maintaining the standards "to attract FDI", the central bank must hinder the Denar from appreciating. The market forces, if the central bank does not intervene, may cause an appreciation of the Denar relative to the Euro. This appreciation would result in an increase in price from P2 to P1 and in effect shift supply from S2 to S1 (or seen by decrease in quantity from Q2 to Q1). To avoid this scenario, the central bank devaluates the Denar by managing foreign reserves. By increasing the foreign reserves in Euros, the central bank sells Denar, hence increasing supply from S1 back to S2. This will likely cause the price to readjust to P2 from P1. Quantity may then increase from Q1 to Q2.

(The removal of import quotas on NM's wine trade is seen in diagram figure 4.)

FIGURE 4. REMOVAL OF QUOTAS ON WINE INDUSTRY IN NM



A quota on wine essentially sets a limit to imports at Q5, at which point imports are stopped and the domestic price at P3 and supply at Q4 are assumed. The text informs however that these quotas must be "removed upon entry into the common market". The removal of the quotas would reduce the supply of wine from domestic producers from a+c+d to only a with the change from S1+QUOTA and P3 to S(WORLD) and P2. The removal would lower domestic producer revenue from f+a+c+d+g+h to only a.

(g) North Macedonia hopes to join the EU, in economic terms the entry into this common market will require and bring many changes. There are many economic benefits but also problems with joining the EU. Overall, the economic benefits of the EU common market outweigh the deficiencies connected with this process.

Economic issues may arise due to EU membership but will likely have a minor impact. An immediate impact of joining the EU, would be losing the power of a currency (Denar) and central bank. The power to adjust the currency will be given to the European central Bank in Frankfurt (as seen in figure 3. The loss of this control will hinder the FDI income, as the value of the currency will adjust to the Euro's value. The text informs that NM has found a lot of success with its adjusting of currency to provoke economic growth. Another disadvantage is seen in the removal of quotas in figure 4, as the producers domestically will lose supply and of revenue drastically. The text also informs that a large part of FDI will be repatriated.

There are more advantages to joining the EU. The article shows that empirically the entry has "a trusted foreign investment" and the inverse FDI "may occur". Other advantages involve the removing of trade barriers. The last points to "free movement of labour." These factors may cause a shift in LRAS from LRAS1 to LRAS2 as seen in figure 1. Increasing Real GDP from Y1 to Y2.

Overall it is evident that joining the EU will likely help the NMs improve economic growth in the long-term.